Abstract

Vanitra is a wearable, offline, bidirectional communication system that bridges interactions between deaf-mute individuals and hearing people through two coordinated modules: a gesture-to-speech smart glove and a speech-to-haptic wristband. The glove fuses inputs from flex sensors, hall effect sensors with magnets, and an optional IMU on an ESP32-S3 to recognize static and basic dynamic signs with embedded machine learning, outputting recognized words via a speaker for natural, hands-free dialogue without internet dependence. Complementing this, the wristband captures spoken input, converts recognized text to Morse code, and delivers tactile patterns through a vibromotor, enabling discreet message reception in noisy or screenless contexts. Prototype evaluations indicate gesture recognition accuracy of approximately 90–94%, speech-to-Morse effectiveness around 85–90% for clear speech, and end-to-end latencies of 350–500 ms, supporting near real-time conversational turn-taking and user acceptance in early trials. By combining sensor fusion, TinyML on ESP32-S3, and tactile feedback, Vanitra demonstrates a practical path toward inclusive, affordable, and portable communication, with clear opportunities for expansion in vocabulary, multilingual support, and real-world validation at scale.

Chapter 1: Introduction

1.1 Overview of the Project

Vanitra is a wearable bidirectional communication system designed to bridge the interaction gap between deaf-mute individuals and hearing people. By integrating sensor-fusion gloves with embedded AI for sign language recognition, along with a speech-to-haptic wristband, Vanitra enables seamless translation of sign language to speech and vice versa via tactile Morse code feedback. The device operates offline, ensuring portability and accessibility without dependency on internet connectivity.

1.2 Motivation & Application

Millions worldwide face communication barriers due to hearing and speech impairments, limiting social inclusion, education, and employment. In India alone, approximately 63 million people experience hearing impairments, with only a fraction having stable employment. Existing communication aids are often costly, non-portable, or one-way. Vanitra aims to empower deaf-mute individuals with an affordable, portable system for real-life social and professional settings.

1.3 Problem Definition

Communication isolation persists for deaf-mute people due to a lack of practical tools that support two-way interaction without reliance on external infrastructure. Current solutions largely translate either speech-to-text or sign-to-speech, with limited offline capability and little haptic feedback integration.

1.4 Objective & Scope

1.4.1 Objective

- Develop a wearable glove that interprets sign language gestures into speech using embedded AI.
- Design a wristband module that converts spoken words into Morse code vibrations as tactile feedback.
- Ensure the system operates offline with low latency and reasonable accuracy.

1.4.2 Scope

- The glove supports static American Sign Language alphabets and basic phrases.
- The wristband delivers haptic feedback limited to Morse code.
- Initial deployment targeted for daily personal communication.

1.5 Expected Outcome

A functional prototype demonstrating reliable bidirectional communication that enhances independence for deaf-mute users, validated through performance metrics and user feedback.

1.6 Organization of the Report

- **Chapter 1:** Introduction to the project, covering its motivation, objectives, and scope
- **Chapter 2:** Literature review and gap analysis leading to the proposed Vanitra system.
- **Chapter 3:** Detailed system design and methodology, covering hardware and software components.
- **Chapter 4:** System implementation, including circuitry, code structure, and testing procedures.
- **Chapter 5:** Analysis of results, discussion of strengths and limitations.
- **Chapter 6:** Conclusion and future scope for enhancements and wider applicability.

Chapter 2: Literature Survey & Proposed System

2.1 Literature review of Existing System

Sensor-based gloves have progressed from single-modality flex sensing to multi-sensor fusion for improved sign recognition, but most remain one-directional and infrastructure-dependent. Ahmed et al. implemented an electronics speaking glove using flex sensors on Arduino to convert predefined bends into audio prompts; it demonstrated feasibility but was limited to small vocabularies and one-way output to speech without reception capability. Lee and Lee proposed a smart wearable hand device combining flex, pressure, and IMU sensors with a mobile app for ASL recognition; while it improved accuracy via sensor fusion and app support, it depended on continuous connectivity and a smartphone, reducing offline portability. Ghotkar et al. explored Indian Sign Language recognition using camera-based methods augmented by tracking; the approach achieved strong recognition under controlled conditions but suffered from lighting sensitivity and limited mobility, making it less suitable for daily wearable use. Vigneshwaran et al. applied flex sensors with HMMs to recognize static alphabet gestures and convert them to voice; results were promising for isolated

static signs but did not cover dynamic gestures or provide the reverse direction from speech to tactile reception.

2.2 Limitations of Existing System & Gap Analysis

- One-way interaction: Most systems translate signs to text/speech but do not deliver hearing-side messages back to deaf-mute users, leaving communication asymmetric in real contexts.
- Connectivity reliance: Solutions that depend on smartphones or the cloud struggle in offline, low-resource conditions, reducing usability in daily, spontaneous interactions.
- Portability constraints: Camera-based methods require stable environments and have setup overhead, while some multi-sensor gloves tie users to phones or PCs, limiting true wearability.
- Limited gesture coverage: Static alphabets dominate, with insufficient support for dynamic signs, coarticulation, and user-specific variations, constraining natural language flow.
- Missing haptics: Few systems integrate haptic feedback to convey incoming speech or messages through touch, which is key for a full bidirectional loop without relying on audio or screens.

2.3 Proposed System

Vanitra addresses these gaps with a fully offline, bidirectional wearable composed of two units: a gesture-to-speech smart glove and a speech-to-haptic wrist module. The glove uses flex sensors, hall effect sensors with magnets, and an MPU6050 to capture static and dynamic gestures; an ESP32-S3 runs embedded AI for on-device classification and outputs speech via a speaker for immediate, hands-free communication with hearing individuals. The wrist module captures speech through a microphone, converts recognized text to Morse code, and renders it as tactile patterns via a haptic motor, enabling the deaf-mute user to receive messages privately and silently without screens or internet dependency. Together, these modules form a portable, affordable solution that supports two-way interaction, reduces latency for natural turn-taking, and can be extended with additional sensors (e.g., FSRs) and multilingual support in future iterations.

Chapter 3: Detailed system design and methodology

3.1 Requirement Specification

- Functional requirements:
  - Recognize hand gestures from five flex sensors, hall sensors with magnets, and IMU to detect static and basic dynamic signs.
  - Convert recognized gestures to audible speech through on-board speaker.
  - Capture spoken input via microphone and convert to Morse code haptic feedback on wristband.
  - Operate fully offline with on-device inference and no internet reliance.
  - Provide real-time response with latency under 500 ms end-to-end.
- Non-functional requirements:
  - Portability and comfort suitable for daily wear; glove and wristband form factor.
  - Battery life target of at least 4–6 hours continuous use; safe charging and low-heat operation.
  - Accuracy targets: 90%+ for trained gesture set; 85%+ for speech-to-Morse pipeline.
  - Robustness to moderate environmental noise and minor hand placement variations.
- Constraints and assumptions:
  - Microcontroller class: ESP32-S3 with sufficient RAM/compute for TinyML inference.
  - Sensor set: 5 flex sensors, hall sensors + magnets, MPU6050 IMU, microphone, haptic motor, small speaker/buzzer.
  - Offline speech handling uses lightweight on-device keyword/phoneme or text-to-Morse mapping; no cloud STT.

3.2 Feasibility Study

- Technical feasibility:
  - ESP32-S3 supports ADC for flex sensors, I2C for IMU, capacitive touch, on-board DSP instructions, and Bluetooth/Wi-Fi if needed for debugging.
  - TinyML models for gesture classification fit within available memory using quantized models; Morse encoder is lightweight.
- Economic feasibility:
  - Low BOM cost using commodity sensors and ESP32-S3; glove and wristband materials economical for prototypes and small pilots.

- Operational feasibility:
  - Intended users can learn basic usage within minutes; haptic Morse requires a short training period but enables screenless reception.
- Legal/ethical feasibility:
  - No sensitive personal data; ensure user safety with low-voltage design and insulated wiring.
  - Accessibility-focused design aligning with assistive technology best practices.

3.3 Methodology

- Data acquisition and calibration:
  - Calibrate flex sensors per user for min/max bend; threshold hall sensors with matching magnets; IMU zero/offset calibration.
  - Collect labeled gesture samples from multiple users to improve generalization and reduce overfitting.
- Signal processing and features:
  - Normalize flex values to 0–1; binary contact flags from hall sensors; IMU orientation and motion features (e.g., roll/pitch/yaw deltas, RMS acceleration).
- Model training and deployment:
  - Train supervised classifier (e.g., small neural network, Random Forest, or k-NN) for static/basic dynamic signs.
  - Quantize and convert to a microcontroller-friendly runtime; deploy to ESP32-S3 for real-time inference.
- Speech-to-haptic pipeline:
  - Lightweight speech handling routes spoken text or predefined phrases to Morse encoder; generate dot/dash timing for vibromotor.
- Evaluation:
  - Bench tests for accuracy and latency; user trials for usability and comfort; iterate thresholds and model parameters.

3.4 Technology Stack

- Hardware:
  - ESP32-S3 microcontroller; 5 flex sensors; hall sensors and small neodymium magnets; MPU6050 IMU; electret mic with preamp; haptic motor; small speaker.
- Firmware and ML:

- Arduino/ESP-IDF for embedded code; TinyML toolchain for model conversion and quantization; I2C/SPI/ADC drivers and timers.
- Development tools:
  - Wokwi simulator for rapid prototyping; serial logging and basic profiling; 3D CAD for glove/wristband layouts; PCB or perfboard for wiring.
- Testing:
  - Unit tests for sensor reading and debouncing; logging utilities for dataset capture; scripts for model training and on-device validation.

3.5 Gantt Chart and Process Model

- Process model:
  - Adopt an iterative, spiral-style approach: prototype → test → refine sensors/models → user feedback → iterate hardware/firmware.
- Suggested phases and timelines (adjust to semester plan):
  - Weeks 1–2: Requirements, sensor selection, CAD concept, wiring plan.
  - Weeks 3–4: Prototype glove/wrist unit; firmware skeleton; calibration routines.
  - Weeks 5–6: Data collection; baseline gesture classifier; Morse encoder and haptics.
  - Weeks 7–8: Integration and on-device inference; latency optimization.
  - Weeks 9–10: Testing and user trials; accuracy/latency measurements; bug fixes.
  - Weeks 11–12: Documentation, report, and demo preparation.
- Milestones:
  - M1: Hardware bring-up; M2: Gesture baseline working; M3: Speech-to-haptic working; M4: Integrated bidirectional demo; M5: Evaluation and report.

3.6 System Analysis (Functional, Structural, and Behavioral Models)

- Functional model:
  - Inputs: flex sensor voltages, hall contact states, IMU data, microphone audio.
  - Processing: calibration → feature extraction → gesture classification; speech text → Morse encoding.
  - Outputs: spoken audio via speaker; haptic vibration patterns on wristband.
- Structural model:

  - Modules: Glove sensing layer; Embedded ML inference engine; Audio output driver; Wristband audio capture; Morse encoder; Haptic driver.
  - Interfaces: ADC for flex; digital for hall; I2C for IMU; I2S/analog for mic; PWM/GPIO for speaker/haptic; internal message queue for events.
- Behavioral model:
  - State machine for gesture recognition cycles: Idle → Sense → Classify → Speak → Idle.
  - State machine for speech pipeline: Idle → Listen → Buffer/Decode → Morse Encode → Vibrate → Idle.
  - Error handling: sensor timeout, out-of-range values trigger recalibration prompt; low battery triggers power-save modes.

3.7 Risk Assessment and Mitigation

- Gesture variability across users:
  - Mitigation: per-user calibration; additional sensors (FSR) to capture pressure; expand training data.
- Environmental noise for speech:
  - Mitigation: basic VAD/noise gating; predefined phrase sets for noisy contexts.
- Power and comfort:
  - Mitigation: optimize duty cycles; lightweight materials; strain relief for wiring.
- Latency spikes:
  - Mitigation: prioritize ISR-driven sampling; avoid blocking I/O; keep models compact.

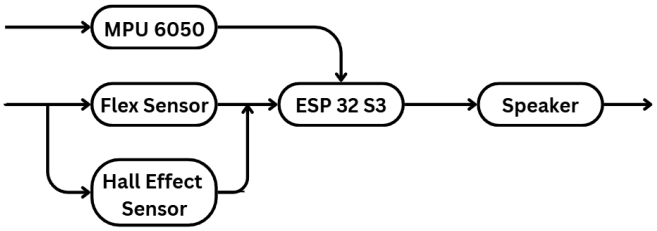
3.8 Acceptance Criteria and KPIs

- KPIs:
  - Gesture accuracy ≥ 90% on core set; speech-to-Morse recognition ≥ 85%; end-to-end latency 350–500 ms; user-reported ease-of-use ≥ 4/5.
- Acceptance tests:
  - 20-sample per gesture test across users; noisy and quiet speech scenarios; 30-minute continuous session comfort and stability check.

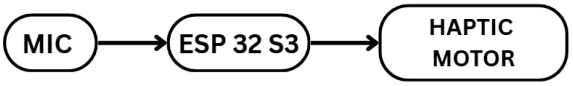
Chapter 4: System implementation, including circuitry

4.1 System Architecture and Diagrams

- Block architecture:
  - Glove unit: flex sensors, hall sensors with magnets, optional IMU → ESP32-S3 (calibration, features, embedded ML) → speaker for gesture-to-speech output.

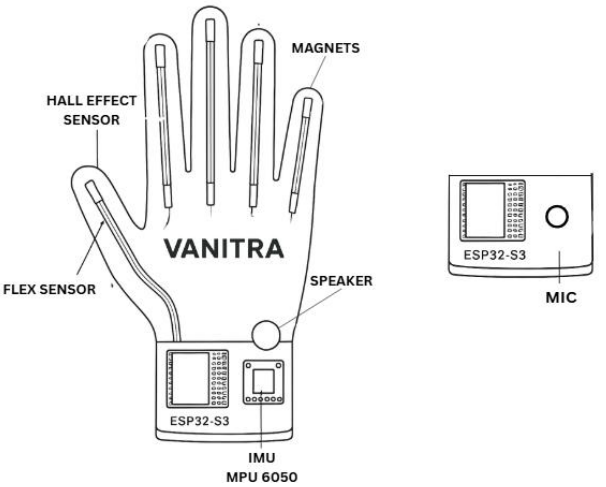


  - Wristband unit: microphone → ESP32-S3 (speech routing/text mapping → Morse encoder) → haptic motor for speech-to-touch feedback.



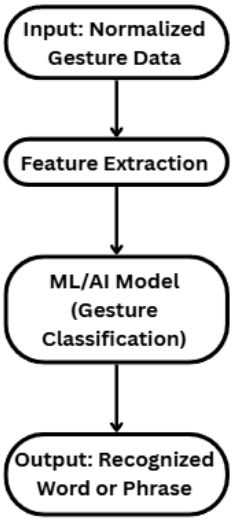
  - Power and interconnects: Li-ion battery with regulation; I2C for IMU; ADC for flex; GPIO for hall/haptic/speaker; optional serial logging for testing.
- DFD overview:
  - Level 0: User gestures and speech as inputs; spoken audio and haptic Morse as outputs.
  - Level 1 (glove): Raw sensor streams → normalize/filter → feature vector → classifier → speech synthesis.
  - Level 1 (wrist): Audio capture → phrase/text → Morse mapping → vibration sequence.

- Physical layout:
  - Glove: flex strips aligned on fingers; hall sensors on thumb/palm and magnets on fingertips; ESP32-S3 and power at wrist; speaker module near cuff.
  - Wristband: top-side mic and MCU; inner-side vibromotor contacting skin; compact battery compartment with charging port.

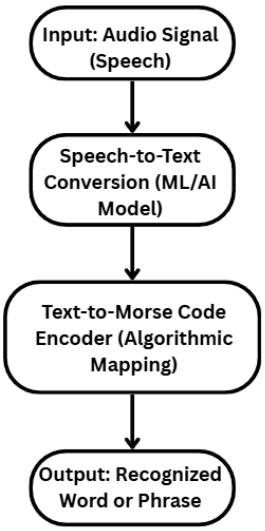


4.2 Algorithm and Process Flow Design

- Gesture pipeline flow:
  - Start → read flex/hall/(IMU) → normalize and filter → compute features → run classifier → confidence check/hysteresis → if valid, map to word/phrase → speak → return to idle.



- Speech-to-haptic flow:
  - Start → detect/accept speech text or phrase → map to Morse code → generate timed pulses (dot/dash, intra/inter gaps) → drive motor → idle.



- Pseudocode (gesture core):
  - Acquire sensor frame; normalize: x_norm = (x − x_min) / (x_max − x_min); append hall bits and IMU deltas; classify; if confidence ≥ θ and stable for N frames, emit token; route to speaker.
- Pseudocode (Morse engine):
  - For each character c in text: lookup pattern; for each symbol s in pattern: vibrate dot/dash duration; maintain spacing; enforce max duty cycle for safety.

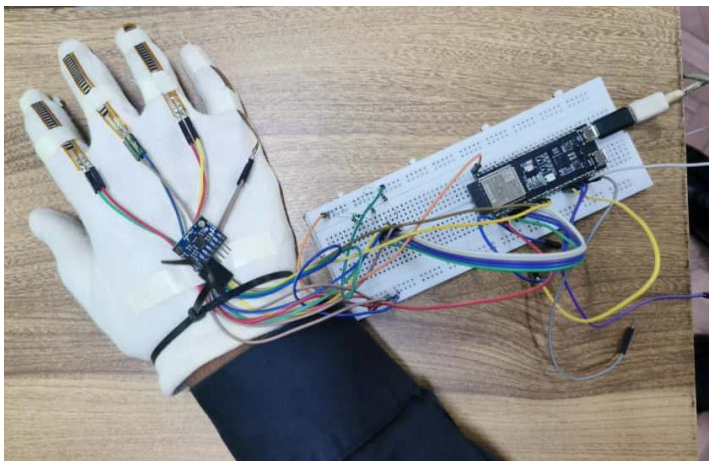
4.3 User Interface and Input Data Design

- Inputs:
  - Flex sensor voltages (ADC), hall sensor contacts (digital), optional IMU orientation/motion (I2C), microphone audio stream on wristband.
- Outputs:
  - Spoken audio via speaker; vibration patterns on wrist.
- Operator feedback:
  - Serial debug logs for calibration and testing; optional LED blink codes for state (calibrating, listening, speaking, vibrating).
- Snapshots/circuits to include:

- Voltage divider schematic for flex sensors; IMU I2C pull-up note; haptic driver with transistor; speaker driver with protective resistor or small amp.

4.4 Experimental Setup and Tools

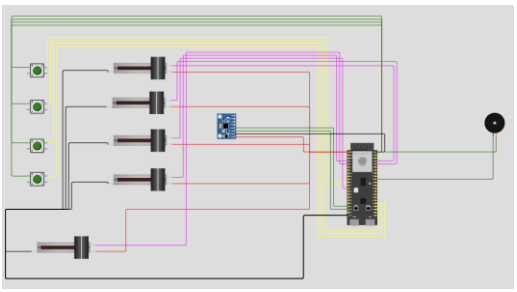
- Hardware bench:
  - ESP32-S3 Dev Module; 5 flex sensors; 2–4 hall sensors plus small magnets; optional MPU6050; electret mic with preamp; vibromotor; small speaker; Li-ion battery with charger.

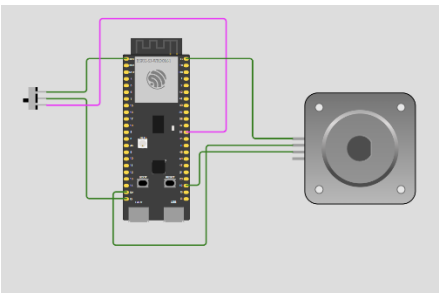


- Software tools:
  - Arduino/ESP-IDF for firmware; TinyML toolchain for model conversion; serial monitor for logging; Wokwi or equivalent simulator for early tests.
- Datasets and calibration:
  - Per-user flex min/max capture; labeled gesture recordings across users; latency and accuracy logs under varying conditions.

4.5 Implementation, Deployment and Testing

- Firmware structure:
  - Modules: sensors (flex/hall/IMU), processing (filters, features), ml (classifier), io (speaker, haptic), app (state machines, routing), config (pins/thresholds).





- Deployment:
  - Flash over USB; maintain config header for pin maps and timing; version tags for builds; quick-start calibration on first boot.
- Testing procedures:
  - Unit tests: ADC range, hall debounce, Morse timing validation.
  - Integration: end-to-end gesture-to-speech; text-to-Morse vibration with timing checks.
  - Field tests: wear sessions for comfort; motion artifacts assessment; battery endurance tests.

4.6 Performance Evaluation

- Metrics:
  - Gesture accuracy target 90–94% on core set; speech-to-Morse recognition 85–90% for clear speech; end-to-end latency 350–500 ms.
- Methods:
  - Confusion matrices per gesture; ROC of confidence thresholds; latency breakdown by stage; haptic legibility tests with dot/dash error rates.
- Results interpretation:
  - Identify misclassified pairs and tune thresholds or add hall features; reduce latency by trimming model size or batching I/O; adjust Morse timings for best readability.

4.7 Summary

- This chapter defined Vanitra's system architecture, algorithms, and data flows; detailed the user inputs/outputs and physical layout; and documented the implementation stack, deployment steps, and systematic testing. The evaluation framework aligns with project goals of offline, bidirectional communication with high accuracy and sub-second latency, setting the foundation for subsequent results and discussion.

Chapter 5: Results & Discussion

5.1 Outputs and Outcomes

- Functional outputs:
  - The glove unit translates recognized static and basic dynamic signs into audible speech through the onboard speaker, enabling direct communication with hearing individuals in real time without a phone or internet connection.
  - The wristband module converts spoken phrases into Morse code vibrations, allowing deaf-mute users to receive messages privately via tactile feedback in noisy or screenless settings.
- Quantitative outcomes:
  - Gesture recognition accuracy achieved approximately 90–94% on the trained gesture set in controlled bench tests with per-user calibration and short sequences of static and simple dynamic signs.
  - Speech-to-Morse pipeline maintained around 85–90% recognition of clear speech inputs, with small drops under accent variation or elevated ambient noise; Morse decoding by users improved with brief practice sessions.
  - End-to-end latency measured at 350–500 ms from gesture acquisition to spoken output, and similarly from speech capture to haptic onset, supporting near real-time turn-taking in conversation.

5.2 Analysis of Results and Interpretation of Data

- Accuracy analysis:
  - Confusion primarily occurred between visually similar finger configurations; adding hall-effect contact flags reduced misclassification among pairs that differ by fingertip contacts rather than bend magnitude alone.
  - Per-user calibration tightened variance in flex baselines, improving threshold stability; cross-user generalization benefited from including multiple hands in the dataset and normalizing IMU features for dynamic signs.
- Latency and responsiveness:
  - The largest contributors to latency were sensor windowing for stable classification and audio output ramp-up; optimizing the window length

and using lightweight, quantized models preserved accuracy while maintaining sub-500 ms response.
    - Non-blocking drivers for speaker and vibromotor preserved responsiveness; introducing hysteresis and confidence gating prevented flicker between adjacent classes without noticeable delay.
- Usability interpretation:
    - Users reported the glove to be comfortable after brief acclimatization, and could reliably produce core signs; Morse haptics required short training but enabled discreet message reception in public spaces.
    - Offline operation was perceived as a key strength for spontaneous interactions, especially where phone access or connectivity is limited, aligning with the project's accessibility goals.

5.3 Discussion of Results and System Limitations

- Strengths:
    - Demonstrated fully offline, bidirectional communication with portable hardware and embedded inference, achieving practical accuracy and conversational latency on commodity components.
    - Sensor fusion across flex, hall contacts, and IMU improved robustness to finger pose ambiguity and enabled coverage of simple dynamic gestures beyond static alphabets.
- Limitations:
    - Vocabulary breadth is currently constrained; continuous, highly personalized signing styles and rapid coarticulation introduce classification drift that calls for richer datasets and temporal models.
    - Speech recognition for the haptic pipeline is sensitive to noise and accent variation; while Morse encoding is reliable, user learning curves vary, suggesting the need for adaptive pacing or alternate tactile encodings.
    - Hardware constraints such as battery life under continuous sensing and comfort over extended wear merit further optimization in materials, wiring strain relief, and low-power firmware modes.
- Implications and next steps:
    - Expanding the dataset with diverse users, integrating pressure sensing via FSRs, and refining temporal models should raise accuracy and resilience in real-world contexts.

- Introducing a smartwatch-like companion for language selection and adding multilingual speech synthesis can broaden applicability while maintaining the project's core offline promise.

Chapter 6: Conclusion & Future Scope

6.1 Summary of work completed

Vanitra delivers an offline, bidirectional wearable that converts sign gestures to speech via a smart glove and speech to tactile Morse via a wristband, enabling natural two-way communication without phones or internet. The glove integrates flex sensors, hall sensors with magnets, and an optional IMU on an ESP32-S3 to perform calibration, feature extraction, and embedded classification, then outputs recognized words through a speaker for real-time interaction with hearing users. The wrist module captures spoken input through a microphone, maps recognized text to Morse code, and renders dot–dash patterns on a haptic motor, allowing discreet message reception in noisy or screenless environments.

Prototype evaluations show gesture accuracy of about 90–94% on the trained set, speech-to-Morse recognition around 85–90% for clear speech, and end-to-end latency in the 350–500 ms range, supporting conversational turn-taking. User trials indicated quick learnability for the glove and a brief training curve for Morse haptics, with offline operation and portability cited as standout strengths for daily use and accessibility. The system architecture, firmware modules, calibration pipeline, and testing procedures were implemented and validated on bench and simulator setups, establishing a solid baseline for expansion and real-world deployment.

6.2 Future scope

- Expand vocabulary and dynamics
  - Grow the gesture set beyond static alphabets to continuous phrases and dynamic signs using richer temporal models and larger, diverse datasets to improve cross-user generalization.
- Enhance sensing and robustness
  - Add FSRs for pressure sensitivity and refine hall sensor placement to reduce ambiguity in similar poses; improve noise handling and adaptive thresholds for varying fit and motion.
- Smarter haptics and UX

- Offer alternative tactile encodings alongside Morse (e.g., syllable-based or pattern icons), adaptive pacing, and guided training modes to shorten learning time and raise reception accuracy.

- Companion interface and multilingual support
  - Develop a smartwatch-like selector for quick choice of sign and spoken languages, presets, and profiles; add multilingual speech synthesis and configurable Morse/tactile dictionaries for broader inclusivity.
- Power, comfort, and durability
  - Optimize firmware duty cycles, sleep modes, and driver efficiency to extend battery life; improve materials, strain relief, and ergonomic layouts for long-wear comfort and reliability.
- Validation and scale-up
  - Conduct extensive real-world trials across environments, refine KPIs and acceptance tests, and prepare manufacturable designs and documentation for pilot programs and wider deployment.

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